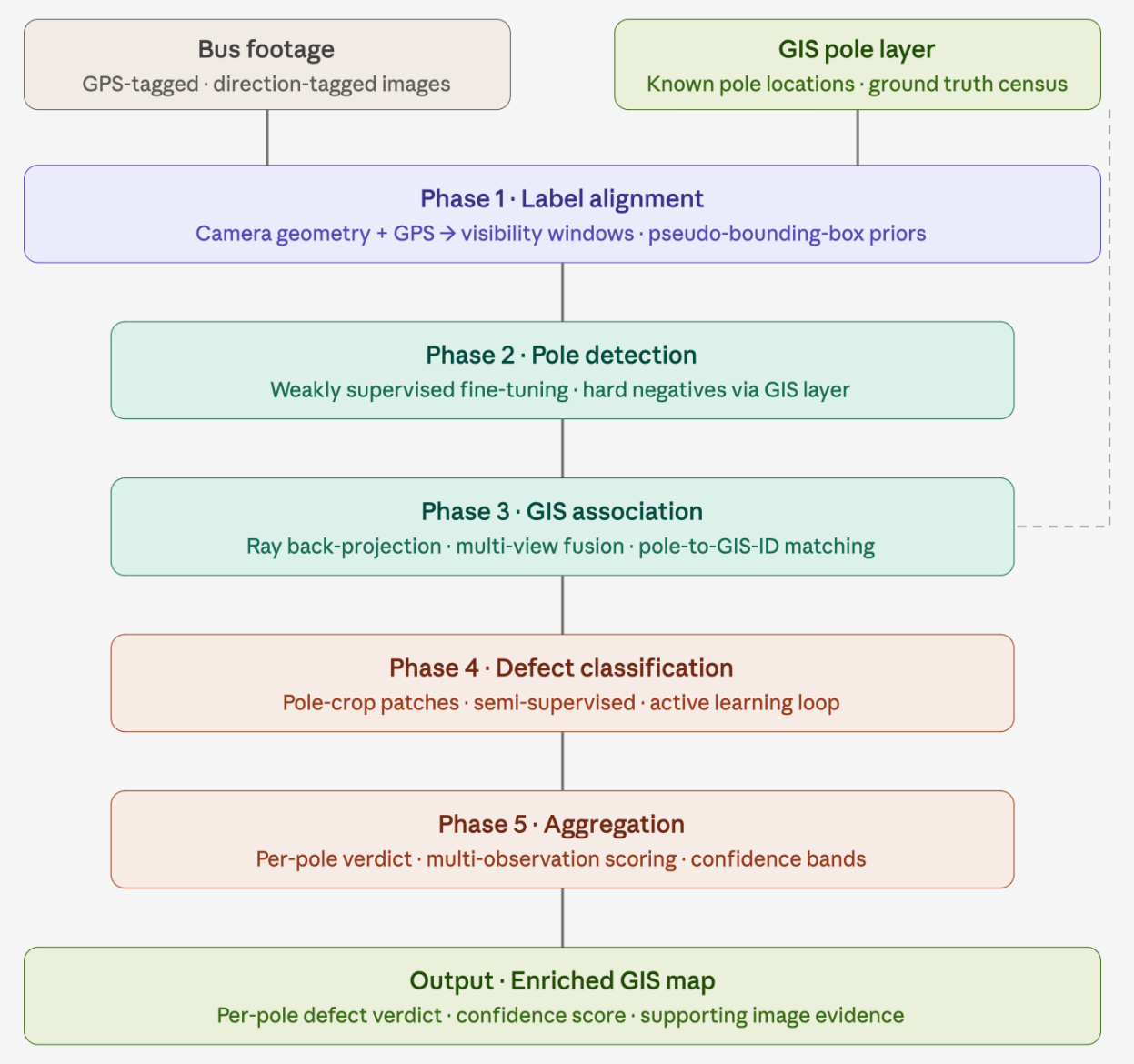
Project Proposal
Phone Pole Condition Assessment
Giles Lemmon

Project Brief:

I have access to time-series images captured from busses in Vermont. I want to build a tool that can take in those images, identify which contain phone poles, assess the condition of those poles, and return a file noting the condition of each one. This should be robust to new images and highly automated.



Phase 1: I will start by making a process to make weak labels through camera geometry to determine when pole is visible
Phase 2: I will use those weak labels to associate images with poles, and train a model to detect the presence of poles in frames.
Phase 3: I will use this model to annotate all images, and get a GIS layer of images and their associated pole
Phase 4: I will then classify the condition of poles based on images. This will likely require manual labeling or assessing the performance of out of the box models for condition assessment.
Phase 5: Turning per-detection issue classifications into pole-level verdicts.

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Phase 0: Background Research

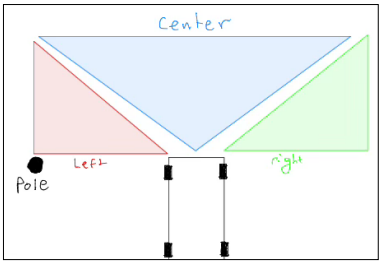
Context:
We need to understand how Electrical Utilities assess phone pole condition, as well as how they like their data presented.

Ongoing Questions:
Is there a standard way of assessing condition of poles?
If this assessment is too large/complex, what are the most important attributes to survey?
In what format do Utilities want this data? GIS layer? If so - what schema?
Currently, how do they do this survey?
How often are poles currently inspected?

Resources Required:
Meeting with someone from a utility (if possible)

Phase 1: Solve the Label Alignment Problem Geometrically

Context:
We have sequences of images, some of which contain phone poles. We want to take the time series of images, and get images of each pole matched to the GPS coordinates of the pole itself. These images are captured by a bus, and are labeled when the bus was closest to the pole. Since the pole is on the side of the bus, it is not present visually in the labeled images.



We have access to:
A) Time-series images captured by a bus every few seconds
B) GIS layers from the state of Vermont with locations of poles
C) Labels for our images corresponding to poles

Solution:
Each label gives the bus GPS at closest approach and the pole's GPS location. Combined with the camera direction tag, I can compute the full geometry of visibility.

At any frame in the sequence, I know the bus position and heading, as well as GPS coordinates of all the poles. I can compute the bearing and distance from the bus to the pole. Given cameras field of view and angle I can figure out if a pole falls within the cameras's view, and if so roughly where it will be.

Output is a valid methodology, used in Phase 2 to generate labels.

Ongoing Questions:
I will be using Field Of View. Are the cameras fixed in place, or do they move?
Are camera positioning standard across busses, or does it change?

Resources Required:
Meeting with someone from a VT Trans to ask about cameras (could be email).

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Phase 2: Pole Detection via Weakly Supervised or Geometry-Guided Training

Context:
I have a sequence of images with no per-image annotations, but I do have geometric pseudo-labels: for each labeled pole, I can compute a window of frames where the pole was visible and an approximate bounding box prior, using the methodology from Phase 1. I will use this to create weak labels.

I have labeled images of phone poles in ideal positions, as well as the full set of images. I need to train a model to detect poles in images. I want to send in all the time series images, and get back the ones with poles in them, as well as the gps location of the poles.

Solution:
I will use a retrained model to identify poles, using the priors (locations) as spatial attention hints. This gives me refined pseudo-bounding-boxes. I now have a weakly labeled detection set.

I will then fine-tune a detector on this domain. Since I have a ground truth of pole locations, I can use this to generate Hard Negatives where I know for certain no pole exists.

Implementation Plan:

1. **Build the projection pipeline** — given a pole GPS + bus pose + camera intrinsics, output predicted (u, v, scale) in image space. Validate this visually on a sample of frames.
2. **Compute visibility windows** — for each labeled pole, walk the frame sequence and flag every frame where the pole falls within the camera frustum. Store as (frame_id, pole_id, predicted_bbox, confidence).
3. **Run pretrained detector in crop mode** — for each visibility window frame, crop the predicted region, run inference, return refined boxes. Filter by score threshold.
4. **Collect hard negatives from GIS** — sample frames where no GIS pole is within any camera frustum. Label these as background.
5. **Fine-tune detector** — train on refined pseudo-positives + hard negatives. Monitor precision using held-out frames where I can manually verify pole presence.
6. **Evaluate and iterate** — Perf metric is recall over the GIS pole set (what fraction of known poles does the detector fire on in at least one frame).

Phase 3: GIS Association and Multi-View Fusion

Context:
Once I have pole detections in image space, I need to associate them with specific GIS entries.

Solution:
Detections that cluster around a GIS point get assigned to that pole. Detections that don't cluster around anything get discarded as noise.

A detection in a frame gives an image-space position (pixel column, scale) plus the known bus position and camera direction. By back-projecting through the camera model, we can estimate a world-space ray or a probability distribution over distance, which intersects with the known GIS pole locations in the vicinity.

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Phase 4: Issue Classification

Context:
Once we have pole detections, we need to assess their condition.

The methodology is still being determined. We can either 1) make labels regarding condition assessment and then train a custom model, or 2) Use out of the box models along with consensus to evaluate condition.

Ideally we can do 2, since it is significantly easier. This would probably involve multiple models to establish a consensus, along with manually annotating a small subset of images to evaluate performance.

Phase 5: Aggregation, Confidence Scoring, and Map Output

Context:
Now it is time to aggregate the condition assessments, score them, and output them in the correct format.

Solution:
The final phase is turning per-detection issue classifications into pole-level verdicts. For each GIS pole, aggregate the defect predictions across all the frames and camera angles that observed it. A pole seen in 12 frames with a lean flag in 9 of them is a high-confidence flag; one with a single ambiguous detection should be suppressed or queued for manual review.